

Functions of many variables – 2

The gradient as an instrument

Homework

11 сентября 2026 года

Theoretical minimum. The mean value theorem survives in $\mathbb{R}^n$ only for real-valued functions; for maps one keeps the inequality $\|f(b) - f(a)\| \leq \sup \|Df\| \cdot \|b - a\|$, and it requires the segment to lie in the domain. A vanishing gradient forces constancy on connected open sets. A Lipschitz gradient gives the descent lemma $|f(y) - f(x) - \langle \nabla f(x), y - x \rangle| \leq \frac{L}{2} \|y - x\|^2$, hence $\|\nabla f(x)\|^2 \leq 2L(f(x) - \inf f)$; convexity is equivalent to the tangent plane lying below the graph and to monotonicity of ∇f ; strong convexity adds a quadratic gap. The coercivity condition $\langle \nabla f(x), x \rangle > 0$ far away replaces compactness. Finally, when f is convex but not differentiable, the gradient is replaced by the subdifferential $\partial f(x)$, and the condition for a minimum becomes $0 \in \partial f(x)$.

1. Prove the descent lemma and deduce that for an L -Lipschitz gradient $\|\nabla f(x)\|^2 \leq 2L(f(x) - \inf f)$. Show that the constant $2L$ cannot be improved.
2. Prove that a differentiable f is convex if and only if ∇f is monotone, and that f is strongly convex with parameter ε if and only if $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 2\varepsilon \|x - y\|^2$.
3. (S.T. Yau CSMC 2016, team) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be convex and C^1 and suppose f has a local minimum x^* .
 - a) Must x^* be a global minimum?
 - b) Consider the gradient method $x_k = x_{k-1} - t \nabla f(x_{k-1})$ with a fixed step $t > 0$. What condition on t guarantees that $f(x_k)$ converges? Prove the convergence.

- c) Assume in addition that f is strongly convex, i.e. $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq m\|x - y\|^2$ for some $m > 0$. Show that (x_k) itself converges.
4. (S.T. Yau CSMC 2018, team) For a convex $f: D \rightarrow \mathbb{R}$ on a convex open $D \subseteq \mathbb{R}^n$, a subgradient of f at x_0 is a vector s with $f(x) - f(x_0) \geq \langle s, x - x_0 \rangle$ for all $x \in D$, and $\partial f(x_0)$ is the set of all of them. Compute $\partial f(0)$ for $f(x) = \|x\|$, and prove that x^* minimises a convex continuous f on $\mathbb{R}^n$ if and only if $0 \in \partial f(x^*)$.
5. Let f be C^1 on a connected open set $U \subset \mathbb{R}^n$ and suppose ∇f takes only finitely many values on U . Prove that f is affine. Does the conclusion survive if ∇f takes countably many values?
6. Prove that if $\langle \nabla f(x), x \rangle > 0$ for all $\|x\| \geq R$, then f attains its global minimum, and that the minimum is attained inside the ball of radius R . Show nevertheless that such an f need not tend to infinity: construct a bounded example with $\langle \nabla f(x), x \rangle > 0$ for all $x \neq 0$.
7. Construct a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ all of whose directional derivatives at the origin exist and depend linearly on the direction, and which is nevertheless not differentiable at the origin; construct another one whose partial derivatives exist everywhere while f is unbounded on every neighbourhood of the origin.
8. Let f be C^2 on $\mathbb{R}^n$ with $|f| \leq 1$ and $\|D^2 f\| \leq 1$ everywhere. Prove that $\|\nabla f\| \leq 2$, and find the best constant for $n = 1$ (compare with Differentiable functions – 4).
9. Let f be C^1 on $\mathbb{R}^n \setminus \{0\}$ with $\langle \nabla f(x), x \rangle = 0$ for all x . Prove that f is constant on every sphere centred at the origin, and that f is determined by a single function of one variable if in addition f is radial.
10. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be C^1 with ∇f nowhere zero. Prove that f has no local extremum and that every level set $\{f = c\}$ is a smooth hypersurface; then decide whether f must be unbounded above.